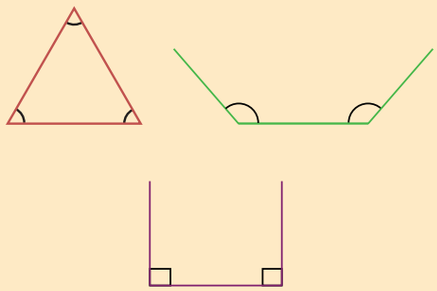
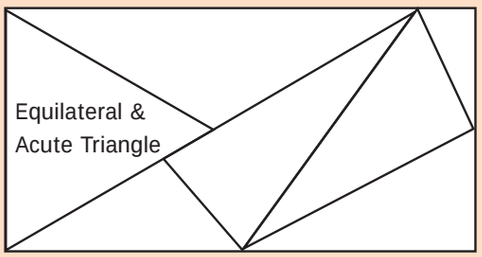
DO YOU KNOW?

A triangle can have three acute angles, but cannot have more than one right angle or an obtuse angle.



Activity

In the given figure, there are some triangles. Measure their sides and angles and name them in two ways. (One is done for you!)



4.3.3 Triangle Inequality property

Think about the situation:

Three students Kamala, Madhan and Sumathi are asked to form triangles with the given sticks of measure **6cm, 8cm, 5cm; 4cm, 10cm, 5cm** and **10cm, 6cm, 4cm** respectively. All of them try to form a triangle. While Kamala, the first girl is successful in forming a triangle, Madhan and Sumathi, next to Kamala are struggling. Why?



When they are trying to join the ends of the two smaller sticks, they find that the two smaller sticks coincide with the longer stick or shorter than the longer stick and they are unable to form triangles. From this, they understand that,

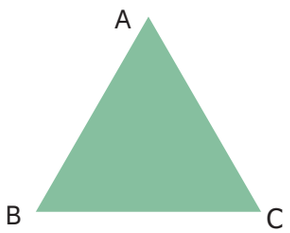
To form a triangle the sum of two smaller sides must be greater than the third side. Thus,

In a triangle, the sum of any two sides of a triangle is greater than the third side. This is known as Triangle Inequality property.

$AB + BC > CA$

$BC + CA > AB$

$CA + AB > BC$



Note

If three sides are equal in length, then definitely a triangle can be formed

DO YOU KNOW?

If any two sides of the triangle are given, then the length of the third side will lie between the difference and sum of the lengths of two given sides.

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